

Question ID: c5e38487

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Medium

Question

A chemist combines water and acetic acid to make a mixture with a volume of **56 milliliters (mL)**. The volume of acetic acid in the mixture is **10 mL**. What is the volume of water, in **mL**, in the mixture? (Assume that the volume of the mixture is the sum of the volumes of water and acetic acid before they were mixed.)

Correct Answer: 46

Rationale

The correct answer is **46**. It's given that a chemist combines water and acetic acid to make a mixture with a volume of **56 milliliters (mL)** and that the volume of acetic acid in the mixture is **10 mL**. Let ***x*** represent the volume of water, in mL, in the mixture. The equation ***x* + 10 = 56** represents this situation. Subtracting **10** from both sides of this equation yields ***x* = 46**. Therefore, the volume of water, in mL, in the mixture is **46**.